

4. Numerical Solution of the Wave Equation [150 points]

The scalar wave equation in d dimensions ($d = 1, 2$) is

$$\frac{\partial^2 u}{\partial t^2} = c^2 \nabla^2 u + S(\mathbf{x}, t), \quad (1)$$

where $u(\mathbf{x}, t)$ is the displacement field, c is the wave speed, and $S(\mathbf{x}, t)$ an external source term. We consider initial conditions

$$u(\mathbf{x}, 0) = u_0(\mathbf{x}), \quad \frac{\partial u}{\partial t}(\mathbf{x}, 0) = v_0(\mathbf{x}), \quad (2)$$

and boundary conditions chosen from:

- Dirichlet: $u = 0$ at domain boundaries,
- Neumann: $\partial_n u = 0$ (no-flux),
- Absorbing (radiation): $\partial_t u + c \partial_n u = 0$ at the edges.

Part A: One-Dimensional Problem: Domain $x \in [0, L]$, uniform grid spacing Δx , time step Δt . Define $u_i^n \approx u(x_i, t^n)$, with $x_i = i \Delta x$, $t^n = n \Delta t$. Use the second-order central-difference (leapfrog) update:

$$u_i^{n+1} = 2u_i^n - u_i^{n-1} + \lambda^2 (u_{i+1}^n - 2u_i^n + u_{i-1}^n) + (\Delta t)^2 S_i^n, \quad (3)$$

where the Courant number is $\lambda = c \Delta t / \Delta x$. Initialize u_i^1 using

$$u_i^1 = u_i^0 + \Delta t v_i^0 + \frac{1}{2} \lambda^2 (u_{i+1}^0 - 2u_i^0 + u_{i-1}^0) + \frac{1}{2} (\Delta t)^2 S_i^0. \quad (4)$$

For stability use the CFL stability condition $\lambda \leq 1$, i.e., including some safety factor, thus something like $\lambda = 0.9$.

For a test use:

$$L = 1.0, \quad c = 1.0, \quad N = 401, \quad \Delta x = \frac{L}{N-1}.$$

Initial condition:

$$u_0(x) = \exp[-(x - x_0)^2 / (2\sigma^2)], \quad v_0(x) = 0,$$

with $x_0 = 0.3L$, $\sigma = 0.02L$.

Run to $T = 1.0$. Compare results for Dirichlet and absorbing boundaries.

Part B: Two-Dimensional Problem: For $u = u(x, y, t)$ on $[0, L] \times [0, L]$:

$$\frac{\partial^2 u}{\partial t^2} = c^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) + S(x, y, t). \quad (5)$$

Discretize with grid points $x_i = i \Delta x$, $y_j = j \Delta y$, $u_{i,j}^n \approx u(x_i, y_j, t^n)$, using equal spacing $\Delta x = \Delta y$. Use the Central-difference Laplacian

$$\nabla^2 u_{i,j}^n = \frac{u_{i+1,j}^n + u_{i-1,j}^n + u_{i,j+1}^n + u_{i,j-1}^n - 4u_{i,j}^n}{\Delta x^2}.$$

and time stepping

$$u_{i,j}^{n+1} = 2u_{i,j}^n - u_{i,j}^{n-1} + (c \Delta t)^2 \nabla^2 u_{i,j}^n + (\Delta t)^2 S_{i,j}^n. \quad (6)$$

The 2D CFL limit is $c \Delta t \leq \Delta x / \sqrt{2}$. Choose $\Delta t = 0.9 \Delta x / (c \sqrt{2})$.

For the actual calculation, choose $L = 1.0$, $c = 1.0$, $N_x = N_y = 201$. Initial condition: a Gaussian centered at $(x_0, y_0) = (0.5L, 0.5L)$:

$$u_0(x, y) = \exp[-((x - x_0)^2 + (y - y_0)^2) / (2\sigma^2)], \quad v_0 = 0.$$

Implement simple Dirichlet and absorbing boundary conditions as options.

Part C: Energy and Accuracy: For both 1D and 2D cases, compute the discrete energy:

$$E^n = \frac{1}{2} \sum_{i,j} \left[\left(\frac{u_{i,j}^{n+1} - u_{i,j}^{n-1}}{2\Delta t} \right)^2 + c^2 \left(\left(\frac{u_{i+1,j}^n - u_{i-1,j}^n}{2\Delta x} \right)^2 + \left(\frac{u_{i,j+1}^n - u_{i,j-1}^n}{2\Delta y} \right)^2 \right) \right] \Delta x \Delta y. \quad (7)$$

For an undamped system ($S = 0$, reflective boundaries), E^n should remain approximately constant. Perform a convergence test: refine Δx and Δt by factor 2 and measure the error at a fixed time to estimate the order of accuracy.

Tasks (In addition to what has already been mentioned above.)

- Implement Python codes for the 1D and 2D problem.
- For 1D, create a space-time color map $u(x, t)$. For 2D, surface plots of $u(x, y, t)$ at various times.
- Show energy $E(t)$ vs. time.
- Plots or animations showing wave propagation.
- Do not forget the report.